Software Applications for Computer Science

Topics you can represent after the midterms

HS Fresenius Cologne - WiSe 26/27 - Marko Duda

Structure of this seminar

1	Friday, September 11, 2026	Quarto/ RStudio/ APA
2	Friday, September 18, 2026	Quarto/ RStudio/ APA
3	Friday, September 25, 2026	Introduction and first hands-on example
4	Friday, October 2, 2026	Operating Systems, basic security and more about CLIs
5	**Friday, October 9, 2026**	**AI ecosystem: LLMs, tokens, agents, harnesses and more**
6	Friday, October 16, 2026	Using Git, Github and Claude for Creating your own Website
7	Friday, October 23, 2026	More Time for Creating your own Website
	———MID TERMS———	After the Mid Terms it is your turn. After the Mid Terms it is your turn. Every week, there will be one to two presentations
8	Friday, November 6, 2026	
9	Friday, November 13, 2026	
10	Friday, November 20, 2026	
11	Friday, November 27, 2026	
12	Friday, December 4, 2026	
13	Friday, December 11, 2026	
14	Friday, December 18, 2026	

Topics for your presentation

1. Git, GitHub and GitLab
2. IDEs
3. Claude Design
4. Figma
5. PowerBI
6. Docker and Containerization
7. Jira for project management
8. Cloud and DevOps

Updated assessment methods and grading criteria

- Almost everybody is doing a presentation alone
- Minimum duration per person is 10min
- Presentation has theoretical and practical part
  - Exception: „Cloud and DevOps“ is only theoretical
  - Practical part: You show how to do a small task on your pc
    - **It is okay if something does not run as expected in practice**
  - Handout is delivered as a markdown file and with terminal commands (if you used some)
- Next slides show you the topics and what you should cover
  - *Optional points to present are marked in italics, decide by the time you have if you want to put them in*
- **You need to upload your handout as Github Flavored Markdown and PDF in Ilias**
  - Handout needs to have between 3 to 5 pages
  - Just google „Markdown to pdf“ for converting your markdown files
  - **Deadline is one day before your presentation**
- **The presentation accounts for 80%, the handout for 20% of the final grade**

Git, GitHub and Markdown

We cover the basics by setting up our own website, you can go more in depth.

- Github
  - Git vs. GitHub: tool vs. platforms
  - Fork, clone, star, watch: finding and reusing other people's code
  - Issues: tracking bugs and ideas
  - Branches and merge/pull requests: how teams review code
  - *Git workflows: feature branch workflow vs. working directly on main*
    - *Understand how multiple people can develop software together*
  - *GitHub Actions: automatic tasks*
- Markdown (the best format for AI)
  - The basics: headings (#), bold/italic, lists, links, images, code blocks, tables
  - Where you see it: README.md, GitHub issues, documentation, your handout
  - Live: create a .md file in GitHub (the handout) and do a merge request

IDEs

- What is an IDE and why not just a text editor
- Components: editor, file tree, terminal, run configurations
- Syntax highlighting, autocompletion, refactoring (e.g. rename)
- Git integration inside the IDE
- Debugger: breakpoints, stepping, variables, call stack (frames)
- Live: set a breakpoint in PyCharm, step through a loop, watch variables change

Claude Design

- Claude Design: generating a design from a prompt, files, website or codebase
- Good prompts: describe purpose, audience, style
- Iterating: fine-grained controls for spacing, color and layout, with changes propagating across the whole design
- Live: Create a design with Claude Design

Claude Design

Additional organizational info, just for Claude Design

- Hint: You need a Claude Pro Subscription (23€ per month) for using Claude Design
- University will refund you one month of the subscription
- Everybody from our seminar will start there Pro subscription on 6th of November
- Claude Desing Presentation will be at November, the 20th
  - You have two weeks for using Claude Design for your presentation

Figma

- Design before code: wireframe, mockup, prototype
- Figma basics: frames, components, auto layout
- *Collaboration: comments, sharing, developer handoff*
- Live: build a simple screen in Figma (frame, component, auto layout)

PowerBI

- What is BI: turning raw data into decisions
- Typical workflow: import data, clean it, model it, visualize it
- Data sources: Excel, CSV, databases
- Visuals and dashboards: choosing the right chart
- Filters and slicers: making dashboards interactive
- *Sharing reports*
- Live: load a CSV and build a small dashboard

Docker and Containerization

- Problem: "it works on my machine"
- Container vs. virtual machine (lightweight, shares the OS kernel)
- Image vs. container (recipe vs. running instance)
- *Docker Hub: downloading ready-made images*
- *Dockerfile: describing your own image*
- Live: run a container (e.g. a web server) and open it in the browser

Jira

For Project Planing; *Italic type is optional*

- Jira: tracking work as tickets (epic, story, task, bug)
- Board, backlog, status workflow (to do, in progress, done)
- Agile (Scrum/Kanban) vs. waterfall: short iterations vs. fixed phases
- Sprint, standup, retrospective in one slide
- *Jira + GitHub: linking tickets to branches and commits*
- Pricing: free plan for small teams (please check current limits before the seminar)
- Live: create a ticket and move it across the board

Cloud and DevOps

No practical part here

- Cloud: renting computers, storage and services instead of buying hardware
  - Service models: IaaS, PaaS, SaaS
  - Big providers: AWS, Azure, Google Cloud
- DevOps: development and operations as one team
  - CI/CD pipeline: code, build, test, deploy automatically
  - Infrastructure as Code: servers described in files
  - Why it is complex: security, costs, scaling, monitoring, many tools interacting
  - Why the profession is well paid: high responsibility and a shortage of experts

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What about doing the presentation together?

- Depending on the number of enrolled students, some topics will be covered by two people together. The list on the right shows the order of priority, with the topics best suited for two presenters first:
  - Each person has a minimum of 10 minutes of presentation time
  - Split the topic between you, so that both of you cover a clear part

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Which topics interest you most? Let me know your first, second and third choice by October 23rd.

Email: marko.duda@ext.hs-fresenius.de